Question - 15: Write a Python program to implement a Decision Tree.

Program Code

```
from math import log2
from collections import Counter


data = [
    (["Sunny", "Hot"], "No"),
    (["Sunny", "Mild"], "No"),
    (["Rainy", "Hot"], "Yes"),
    (["Rainy", "Mild"], "Yes"),
    (["Cloudy", "Hot"], "Yes"),
    (["Cloudy", "Mild"], "Yes")
]

features = ["Weather", "Temperature"]


def entropy(labels):
    total = len(labels)
    counts = Counter(labels)

    return -sum(
        (count / total) * log2(count / total)
        for count in counts.values()
    )


def information_gain(data, feature_index):
    total_entropy = entropy([row[1] for row in data])
```

```
        values = set(row[0][feature_index] for row in data)

        weighted_entropy = 0

        for value in values:
            subset = [row for row in data
                      if row[0][feature_index] == value]

            weighted_entropy += (
                len(subset) / len(data)
            ) * entropy([row[1] for row in subset])

        return total_entropy - weighted_entropy


    def build_tree(data, available_features):
        labels = [row[1] for row in data]

        if len(set(labels)) == 1:
            return labels[0]

        if not available_features:
            return Counter(labels).most_common(1)[0][0]

        best_feature = max(
            available_features,
            key=lambda index: information_gain(data, index)
        )

        tree = {features[best_feature]: {}}

        for value in set(row[0][best_feature] for row in
data):
```

```
            subset = [row for row in data
                      if row[0][best_feature] == value]

            remaining = [
                index for index in available_features
                if index != best_feature
            ]

            tree[features[best_feature]][value] = build_tree(
                subset, remaining
            )

        return tree


    tree = build_tree(data, list(range(len(features))))

    print("Decision Tree:")
    print(tree)
```

Program Output

```
PS E:\Desktop\SIMATS\Artificial Intelligence\AI LAB> & C:/Users/
/Local/Programs/Python/Python314/python.exe "e:/Desktop/SIMATS/A
elligence/AI LAB/experiment15_decision_tree.py"
Decision Tree:
{'Weather': {'Cloudy': 'Yes', 'Rainy': 'Yes', 'Sunny': 'No'}}
PS E:\Desktop\SIMATS\Artificial Intelligence\AI LAB>
```